

1. Aim

To design and implement a VLAN-based enterprise LAN in Cisco Packet Tracer, separating Administration, Faculty, Students, and Guests into different VLANs and configuring router-on-a-stick inter-VLAN routing.

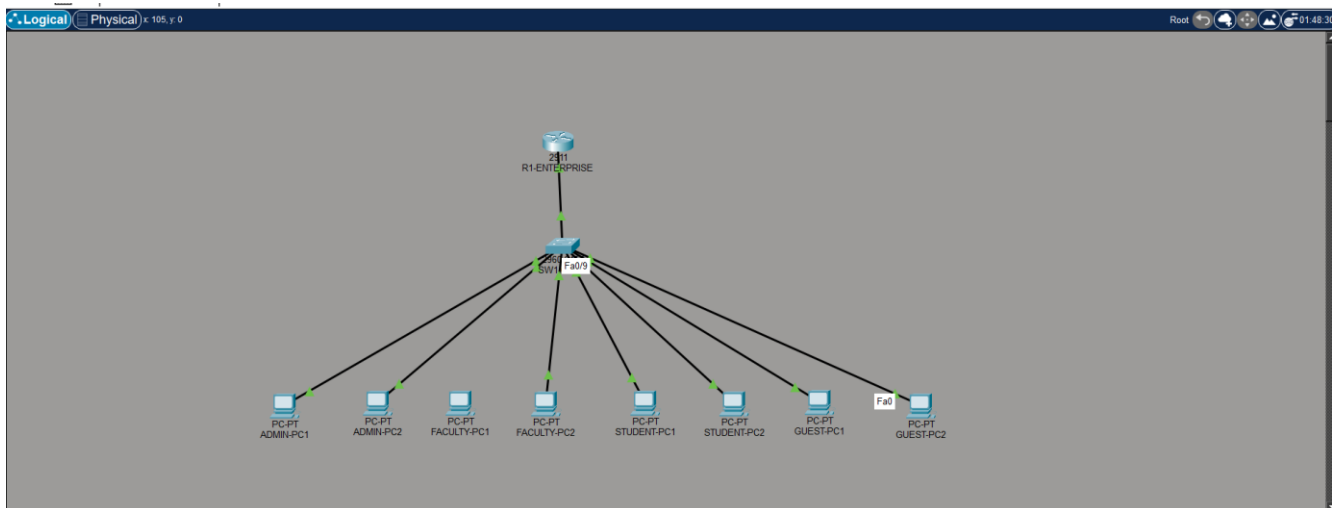
2. Objectives

- Create separate VLANs for Administration, Faculty, Students, and Guests.
- Assign end devices to the appropriate access VLANs.
- Configure an 802.1Q trunk between the core switch and the enterprise router.
- Configure router subinterfaces as the default gateways for the VLANs.
- Verify VLAN membership, trunk operation, IP addressing, and connectivity.
- Document implementation evidence and troubleshooting observations.

3. Network Topology

The implemented topology contains one Cisco 2911 router (R1-ENTERPRISE), one core switch (SW1-CORE), and eight PCs. The router connects to the switch through G0/0 on the router and Fa0/24 on the switch. End devices are grouped into four departmental VLANs.

Figure 1 – Final Network Topology



4. VLAN and Addressing Plan

VLAN ID	Department	Switch Ports	Gateway	Example Host
10	ADMINISTRATION	Fa0/1, Fa0/2	192.168.10.1	192.168.10.10 / .11
20	FACULTY	Fa0/3, Fa0/4	192.168.20.1	192.168.20.x
30	STUDENTS	Fa0/5, Fa0/6	192.168.30.1	192.168.30.x
40	GUESTS	Fa0/7, Fa0/8	192.168.40.1	192.168.40.x

Verified switch VLAN membership: VLAN 10 contains Fa0/1 and Fa0/2; VLAN 20 contains Fa0/3 and Fa0/4; VLAN 30 contains Fa0/5 and Fa0/6; VLAN 40 contains Fa0/7 and Fa0/8.

Figure 2 – VLAN Verification

```
Switch>show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Gig0/1
                                           Gig0/2
10   ADMINISTRATION        active    Fa0/1, Fa0/2
20   FACULTY                active    Fa0/3, Fa0/4
30   STUDENTS               active    Fa0/5, Fa0/6
40   GUESTS                 active    Fa0/7, Fa0/8
1002 fddi-default         active
1003 token-ring-default   active
1004 fddinet-default       active
1005 trnet-default         active
Switch>
```

5.1 Create VLANs on SW1-CORE

The following VLANs were created and named:

enable

configure terminal

vlan 10name ADMINISTRATION

exit

vlan 20

name

```
exit
vlan
name
exit
vlan

40
name
exit
```

5.2 Assign Access Ports

```
interface range fa0/1-2
switchport mode access
switchport access vlan 10
exit
interface range fa0/3-4
switchport mode access
switchport access vlan 20
exit
interface range fa0/5-6
switchport mode access
switchport access vlan 30
exit
interface range fa0/7-8
switchport mode access
switchport access vlan 40
exit
```

5.3 Configure the Router-Switch Trunk

```
interface fa0/24
switchport mode trunk
no shutdown
end
```

The trunk was verified as operational using `show interfaces trunk`. The observed encapsulation was IEEE 802.1Q, the status was trunking, and VLANs 1, 10, 20, 30, and 40 were shown as active on the trunk.

Figure 3 – 802.1Q Trunk Verification

```
Switch>show interfaces trunk
Port      Mode           Encapsulation  Status        Native vlan
Fa0/24    on             802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/24    1-1005

Port      Vlans allowed and active in management domain
Fa0/24    1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    1,10,20,30,40
```

5.4 Configure Router-on-a-Stick Subinterfaces

```
interface gigabitEthernet0/0
no shutdown
exit
interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
interface gigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit
interface gigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
exit
```

The router subinterfaces were verified as up/up with the configured gateway addresses for VLANs 10, 20, 30, and 40.

Figure 4 – Router Sub interface Verification

Switch>show vlan brief			
VLAN	Name	Status	Ports
1	default	active	Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Gig0/1 Gig0/2
10	ADMINISTRATION	active	Fa0/1, Fa0/2
20	FACULTY	active	Fa0/3, Fa0/4
30	STUDENTS	active	Fa0/5, Fa0/6
40	GUESTS	active	Fa0/7, Fa0/8
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	
Switch>			

6. End Device Configuration

Static IPv4 addressing was used for the demonstrated Administration PCs. ADMIN-PC1 was configured with 192.168.10.10/24 and gateway 192.168.10.1. ADMIN-PC2 was configured with 192.168.10.11/24 and gateway 192.168.10.1.

Device	IPv4 Address	Subnet Mask	Default Gateway
ADMIN-PC1	192.168.10.10	255.255.255.0	192.168.10.1
ADMIN-PC2	192.168.10.11	255.255.255.0	192.168.10.1

7. Implementation & Results

The implementation was verified progressively. The VLAN table confirmed the departmental port assignments, the trunk verification confirmed IEEE 802.1Q trunking on Fa0/24, and the router interface summary confirmed the configured VLAN gateway subinterfaces.

7.1 Successful Connectivity Test

ADMIN-PC1 successfully reached the Administration VLAN gateway at 192.168.10.1. The recorded test showed 4 packets sent, 4 received, and 0% packet loss, with an average round-trip time of 5 ms.

7.2 Verification Summary

Verification Item	Observed Result	Status
VLAN creation and naming	VLANs 10, 20, 30, 40 present	Verified
Access-port assignment	Fa0/1–Fa0/8 grouped by department	Verified
Router–switch trunk	Fa0/24, 802.1Q, trunking	Verified
Router gateways	192.168.10.1 / .20.1 / .30.1 / .40.1	Verified
ADMIN-PC1 gateway reachability	4/4 received, 0% loss	Verified
ADMIN-PC2 gateway reachability	4/4 lost during testing	Unresolved

8. Troubleshooting & Analysis

During validation, ADMIN-PC2 could not reach 192.168.10.1 even though its displayed IP configuration was 192.168.10.11/24 with gateway 192.168.10.1. Troubleshooting included checking the switch interface state, VLAN membership, the trunk, and the switch MAC-address table.

- Fa0/24 was initially found down/disabled; it was enabled with no shutdown.
- The router-to-switch physical connection was identified through CDP and corrected to use SW1-CORE Fa0/24.
- Fa0/24 was configured as an 802.1Q trunk and verified as trunking.
- VLAN membership was verified: Fa0/1–Fa0/2 in VLAN 10, Fa0/3–Fa0/4 in VLAN 20, Fa0/5–Fa0/6 in VLAN 30, and Fa0/7–Fa0/8 in VLAN 40.
- ADMIN-PC2 still showed 100% packet loss to 192.168.10.1 during the final recorded test, so this result is reported honestly rather than marked as successful.

This troubleshooting sequence demonstrates analysis of physical status, VLAN membership, trunk operation, addressing, and connectivity rather than relying only on the visual topology.

9. Conclusion

A VLAN-based enterprise LAN was implemented in Cisco Packet Tracer using four departmental VLANs and router-on-a-stick inter-VLAN routing. The VLAN structure, access-port assignments, 802.1Q trunk, and router gateway subinterfaces were configured and verified. ADMIN-PC1 successfully reached its VLAN gateway with 0% packet loss. ADMIN-PC2 remained unresolved in the recorded testing and is documented as a troubleshooting observation.

10. GitHub Upload

After final validation, upload the Packet Tracer project file and the completed report to the assigned GitHub repository. The repository should contain the .pkt project and supporting documentation/screenshots as required by the assignment.

Suggested project file: VLAN_Enterprise_LAN.pkt

11. Screenshot Checklist for Submission

- Final Packet Tracer topology.
- show vlan brief – VLAN names and access-port assignments.
- show interfaces trunk – Fa0/24, 802.1Q, trunking, active VLANs.
- show ip interface brief – router subinterfaces and gateway addresses.